

LAB 2 - Kotlin & Android Studio Basics

May 19, 2026 4:29 PM

Weekly Assignment and Lab Exercise

Assignments

No assignment this week.

Lab Exercise

1. Test all Kotlin code examples from the lecture slides.
2. Complete the Lab Session Week 2:



Lab Session Week 2 F24.docx

1.

COMP-304: Mobile Apps Development



Lab Session Week 2 F24

Lab Session Week 2

In this lab session exercise, you will:

1. Review the basics of Kotlin programming language.
2. Set up Android Studio to use Kotlin and build apps.
3. Develop Android apps using Kotlin.

Exercise 1: Review the Basics of Kotlin

In this exercise, you will review the basics of the Kotlin programming language. Please follow the steps below:

1. Variables

- Define and initialize variables using `val` and `var`.
- Understand the difference between mutable (`var`) and immutable (`val`) variables.
- Example:

```
val a: Int = 1 // immediate assignment
val b = 2      // `Int` type is inferred
var x = 5      // `Int` type is inferred
x += 1
```

2. Control Structures

- Practice using `if`, `when`, `for`, and `while` loops.
- Example:

```
fun maxOf(a: Int, b: Int): Int {
    return if (a > b) a else b
}

for (i in 1..5) {
    println(i)
}
```

3. Functions

- Define and use functions with and without return types.
- Example:

```
fun sum(a: Int, b: Int): Int {
    return a + b
}

fun printSum(a: Int, b: Int) {
    println("sum of $a and $b is ${a + b}")
}
```

4. Classes

- Create simple classes and understand their structure.
- Example:

```
class Person(val firstName: String, val lastName: String, var age: Int)
```

5. Collections

- Work with collections such as lists and maps.
- Example:

```
val items = listOf("apple", "banana", "kiwi")
for (item in items) {
    println(item)
}
```

Exercise 2: Set Up Android Studio to Use Kotlin

In this task, you will set up Android Studio to use Kotlin for developing Android apps. Please follow the steps below:

1. Create a New Project

- Open Android Studio and create a new project.
- Select Empty Activity and ensure that the language is set to Kotlin.

Exercise 3: Develop Android Apps Using Kotlin

In this task, you will develop a simple Android application using Kotlin. Please follow the steps below:

1. Create a New Android Project

- Open Android Studio and create a new project.
- Name the project KotlinBasicsApp.
- Select Empty Activity and set the language to Kotlin.

2. Write Code to Create a Simple Android Application

- Open MainActivity.kt and write the following code to display "Hello, Kotlin!" on the screen:

```
package com.example.kotlinbasicsapp

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp
import com.example.kotlinbasicsapp.ui.theme.KotlinBasicsAppTheme

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContent {
            KotlinBasicsAppTheme {
                Surface(
                    modifier = Modifier.fillMaxSize(),
                    color = MaterialTheme.colorScheme.background
                ) {
                    Greeting("Kotlin")
                }
            }
        }
    }
}
```

```
        }
    }
}

@Composable
fun Greeting(name: String) {
    Column(modifier = Modifier.padding(16.dp)) {
        Text(text = "Hello, $name!")
    }
}
```

3. Run the Application

- Connect an Android Virtual Device (AVD) or a physical Android device.
- Run the application and verify that it displays "Hello, Kotlin!" on the screen.

4. Add Interactive Elements

- Modify the code to include a button that changes the displayed text when clicked:

```
@Composable
fun Greeting(name: String) {
    var text by remember { mutableStateOf("Hello, $name!") }
    Column(modifier = Modifier.padding(16.dp)) {
        Text(text = text)
        Spacer(modifier = Modifier.height(8.dp))
        Button(onClick = { text = "Hello, Android!" }) {
            Text("Click Me")
        }
    }
}
```

Demonstration

Please demonstrate the exercises at the end of the lab session. Ensure that your application runs successfully on an Android device or emulator and that all functionalities are working as expected.

Note: Use the references provided in the Week 2 lecture slides to help you with Kotlin syntax and Android development.